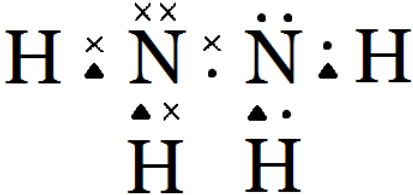


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		furthest line to the left	1			1		
		(ii)		electron falls from higher energy levels to lower energy levels / to n = 2 (1) the difference between any two energy levels is fixed / energy levels are quantised (1)	2			2		
	(b)	(i)		this is the energy required to remove an electron from an atom of hydrogen / for an electron to go from n = 1 to n = ∞ (1) in the gaseous state (1) [award (1) for correct equation with state symbols but no explanation]	2			2		
		(ii)		E = hf (1) f = $2.18 \times 10^{-18} / 6.63 \times 10^{-34}$ (1) f = 3.29×10^{15} (1)	1	2		3	3	
	(c)			V ₂ = $\frac{163 \times 273}{398}$ = 112 (1) n hydrazine = 112/22400 = 0.00500 (1) M _r hydrazine = 0.160/0.005 = 32 so molecular formula is N ₂ H ₄ (1) ecf possible credit alternative method using pV = nRT		2	1	3	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		 shared pair of electrons between N atoms (1) rest of electrons correct (1) accept all electrons represented as dots/crosses						
		(ii)		a covalent bond where the electrons are not shared equally between the atoms / unequal electron density	1			1		
	(e)			$\text{N}_2\text{H}_4 + \text{H}_2\text{O} \rightleftharpoons \text{N}_2\text{H}_5^+ + \text{OH}^-$ accept $\text{N}_2\text{H}_4 + 2\text{H}_2\text{O} \rightleftharpoons \text{N}_2\text{H}_6^{2+} + 2\text{OH}^-$		1		1		
				Question 8 total	7	7	1	15	5	0